

NPCC–NY Lite Weekly Update

Overvoltage repair, downstate delivery spine, generation alignment, and current tie-line mismatch

NPCC / NY-lite Calibration Workstream

July 8, 2026

This Week's Message

Status

The model moved from an overvoltage / ACOPF-feasibility diagnosis to a working S4 calibration candidate.

- The earlier public-load ACOPF failure was not mainly a branch-rating problem.
- Relaxed-voltage solves exposed a **system-wide high-voltage escape path**, with all NY-only buses near 2.0 pu.
- The fix was not one more line alone: it required downstate delivery paths plus missing downstate active-power controllability.
- S3 restored feasibility with a bounded Zone J supply equivalent.
- S4 aligned zonal generation participation using Gold Book capability targets and improved interface behavior.
- Remaining problem: **tie-line / interface residuals are still material**, especially Total East.

Original Failure Mode

What failed

- Public NYISO load shapes were scaled to the preserved NPCC New York load level.
- Normal ACOPF failed under public downstate load relocation.
- Removing ratings alone did not restore ACOPF feasibility.

What the relaxed solve revealed

- ACOPF could solve only after voltage bounds were relaxed to 0.1–2.0 pu and ratings were removed.
- The optimizer drove the whole NY-only subsystem to abnormal high voltage.
- That made the relaxed solution a diagnostic, not a valid operating point.

The core issue became: **AC voltage / controllability infeasibility under public load relocation.**

Overvoltage Evidence

Public scenario	$V_{\min}$	$V_{\max}$	Buses above 1.10 pu
Summer peak	1.9534	2.0000	46 / 46
Winter peak	1.9483	2.0000	46 / 46
Shoulder light load	1.9449	2.0000	46 / 46
High NYC / LI load	1.9534	2.0000	46 / 46
High Total East	1.9453	2.0000	46 / 46
Low Total East	1.9478	2.0000	46 / 46

Interpretation

The overvoltage was not localized. It affected every retained NY bus in the relaxed diagnostic cases, including upstate, lower-Hudson, NYC proxy, and Long Island proxy buses.

Main Causes of the Overvoltage Problem

❶ **Public load relocation changed nodal stress.**

Several GW moved toward G/J/K-style downstate proxy areas that had little or no original NPCC load.

❷ **The reduced topology lacked downstate delivery structure.**

Initial paths could not represent the Central East → Total East → UPNY–ConEd → NYC / LI corridor sequence.

❸ **The boundary equivalent was exact only at S1.**

Fixed measured boundary injections reproduced S1, but did not adapt like a dynamic external network under new public scenarios.

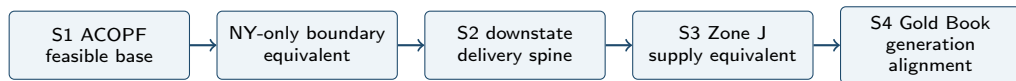
❹ **Voltage and reactive controls were too idealized.**

Generator Q , boundary generator Q , branch charging, and unconstrained voltage objectives allowed abnormal voltage behavior when bounds were relaxed.

❺ **Missing local active-power controllability in Zone J.**

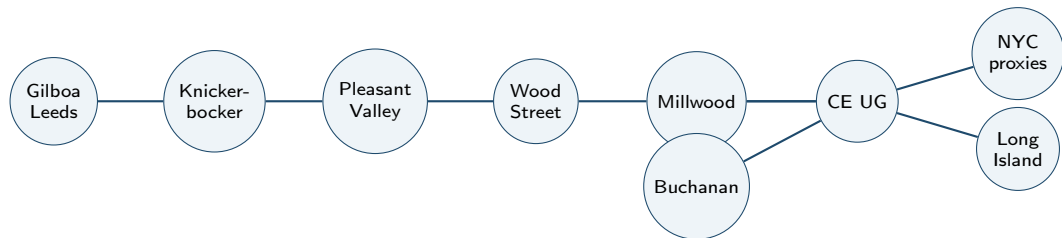
The reduced model was forcing NYC public load to be served through coarse import paths with too little local equivalent supply.

Repair Strategy



- S1: replace embedded infeasible dispatch with an ACOPF-feasible base point.
- NY-only equivalent: remove external NPCC branch dominance while preserving measured S1 boundary behavior.
- S2: add no-load downstate transit paths with zero added branch charging initially.
- S3: add bounded aggregate NYC / Zone J active supply.
- S4: align zonal capability and participation to public NYISO Gold Book patterns.

S2: Downstate Delivery Spine



- Added zero-load transit buses: Knickerbocker, Wood Street, East Garden City.
- Added delivery paths toward Pleasant Valley, Millwood, CE UG, NYC proxies, and Northport.
- Initial new-line parameters used high ratings and $B_c = 0$ to avoid adding more capacitive charging.
- Result: improved public active-load continuation boundary, but did **not** make full public relocation feasible by itself.

S3: Missing Zone J Active-Power Equivalent

Why it was needed

- Public load relocation placed several GW into NYC proxy buses.
- Real NYISO Zone J has substantial local capability.
- The reduced NPCC mapping had proxy buses but lacked sufficient local controllable active supply.

Implementation

- Added bounded aggregate supply at RAV A-3, AK-3, and Goethals.
- Interpreted as NYC local generation / import equivalent, not unit-level modeling.
- Default working cap: 3.0 GW.

Zone J PMAX test	Cases	Successes	Max Zone J dispatch
No Zone J equivalent	6	0	0 MW
2.40 GW threshold	6	5	2400 MW
2.45 GW threshold	6	6	2431 MW
3.00 GW default	6	6	2430 MW

Voltage Issue After S3

- S3 restored normal-bound ACOPF feasibility for public P-only, public Q-only, public P+Q, and public P+Q plus public external schedule tests.
- The 3.0 GW Zone J equivalent was not binding in the broad S3 validation set.
- The previous 2.0 pu escape was eliminated from the accepted operating cases.
- Remaining voltage issue is now milder but still visible: many S4 OPFs sit on 0.90 / 1.10 pu bounds.

Updated interpretation

S3 solved the **feasibility and abnormal-overvoltage** problem, but did not yet calibrate the interface-flow behavior.

Gold Book Generation Alignment

Zone	2025 Gold Book summer capability	Modeling implication
A – WEST	3492.4 MW	Existing NPCC/S3 participation was too high; cap participation.
B – GENESE	780.5 MW	Existing NPCC/S3 participation was too high; cap participation.
C – CENTRL	6781.1 MW	Large resource zone, but still avoid excessive upstream export.
F – CAPITL	4775.6 MW	Add / enable Capital-region equivalent capability.
G – HUD VL	4704.0 MW	Add Hudson Valley equivalent capability near lower-Hudson proxies.
I – DUNWOD	0.0 MW	Reclassify CE UG as interface / voltage support, not native Zone I generation.
J – N.Y.C.	8704.7 MW	Keep bounded Zone J aggregate supply equivalent.
K – LONGIL	5195.5 MW	Add Long Island equivalent capability near Northport / East Garden City.

Source values are from the local project Gold Book target table used for S4 generation alignment.

S4 Model Changes

Added capability

- E: Edic / Porter equivalent.
- F: Gilboa, Albany, Rotterdam, New Scotland equivalents.
- G: Pleasant Valley, Ramapo, Leeds equivalents
- K: Northport and East Garden City equivalents

Participation controls

- A/B/C participation caps, best candidates use factor 0.8.
- CE UG generator moved away from native Zone I role.
- Relative costs tune Zone J vs non-J equivalent participation.
- Spine X sensitivity was secondary to participation alignment.

Final S4 candidate cases reload as **143 buses, 246 branches, 62 generators.**

Optimization Problem: Inner ACOPF

For each public scenario s and candidate parameter set θ , the solved problem is the MATPOWER ACOPF:

$$\begin{aligned} \min_{x_s} \quad & \sum_{g \in \mathcal{G}} \left(c_{2g} P_{g,s}^2 + c_{1g} P_{g,s} + c_{0g} \right) + \rho_V \sum_{i \in \mathcal{B}} (V_{i,s} - V_i^{ref})^2 \\ \text{s.t.} \quad & g_i^P(\theta_s, V_s, P_s, Q_s) = 0, \quad g_i^Q(\theta_s, V_s, P_s, Q_s) = 0 \\ & P_g^{min} \leq P_{g,s} \leq P_g^{max}, \quad Q_g^{min} \leq Q_{g,s} \leq Q_g^{max} \\ & V_i^{min} \leq V_{i,s} \leq V_i^{max} \\ & |S_{\ell,s}^{from}| \leq RATE_{A,\ell}, \quad |S_{\ell,s}^{to}| \leq RATE_{A,\ell}. \end{aligned}$$

- Public P-58C load and P-32 external schedules are fixed inputs for scenario s .
- S3/S4 equivalent-resource costs and limits enter through c_g , P_g^{min} , and P_g^{max} .
- Interface-flow matching is **not** embedded in this inner ACOPF yet.

Outer Calibration Problem

The calibration variables are candidate parameters rather than bus voltages directly:

$$\theta = \{c_{1,J}, c_{1,E}, c_{1,F}, c_{1,G}, c_{1,K}, \alpha_{ABC}, X_{\text{spine}}, P_{J,eq}^{max}\}.$$

Current implemented workflow

$\theta \rightarrow$ build S4 candidate cases $\rightarrow$ solve IPOPT ACOPF for all scenarios $\rightarrow$ score interface residuals.

$$\theta^* = \arg \min_{\theta \in \Theta} J_{\text{cal}}(\theta)$$

$$J_{\text{cal}}(\theta) = \sum_{s \in \mathcal{S}} J_{\text{int}}(\theta; s) \quad \text{subject to OPF success and no branch-overload rejection.}$$

- The grid search tested 178 candidate settings and 1,068 IPOPT ACOPFs.
- All reported S4 candidate objective values are **post-OPF interface scores**, not the raw MATPOWER generation-cost objective.

Interface Objective Used for Candidate Ranking

For each scenario s and monitored interface m :

$$e_{s,m}(\theta) = F_{s,m}^{lite}(\theta) - F_{s,m}^{target}.$$

Classic score

$$J_{\text{classic}}(\theta) = \sum_{s,m} \left[\frac{e_{s,m}(\theta)}{\max(100, |F_{s,m}^{target}|)} \right]^2.$$

Fixed-scale score

$$J_{\text{fixed}}(\theta) = \sum_{s,m} \left[\frac{e_{s,m}(\theta)}{S_m^{\text{fixed}}} \right]^2.$$

$$S_m^{\text{fixed}} = \max(500, \tilde{L}_m^{P32, \text{scaled}}).$$

- The fixed-scale score uses one stable scale per interface from the median available scaled P-32 limit.
- It avoids over-weighting low-flow or missing-limit rows with an artificial 500 MW fallback.
- The current v2 candidate is selected by J_{fixed} .

S4 Calibration Results

Objective values below are post-OPF interface scores accumulated across six public scenarios.

Case	Success	Classic obj.	Mean Zone J eq.	Mean S4 non-J eq.
S3 baseline, 3.0 GW Zone J	6 / 6	893.20	1443 MW	0 MW
S4 equivalents only	6 / 6	910.11	0 MW	2035 MW
S4 equivalents + A/B/C caps	6 / 6	312.50	0 MW	2767 MW
Cost sweep v1, classic-best style	6 / 6	93.52	2861 MW	649 MW
Cost sweep v2, fixed-scale best	6 / 6	132.03	1392 MW	2151 MW

Key point

Generation participation and cost alignment produced the largest improvement. Topology-only changes were necessary, but not sufficient.

Candidate Choice Depends on Scoring Basis

Both columns are post-OPF residual objectives: J_{classic} and J_{fixed} , not separate ACOPF objective functions.

Candidate	Classic obj.	Fixed-scale obj.	Mean Zone J	Mean S4 non-J
v1: J220 / EFGK260 / CAP0.8 / X1	93.52	7.728	2861 MW	649 MW
v2: J180 / EFGK180 / CAP0.8 / X1	132.03	4.335	1392 MW	2151 MW

- v1 remains the better classic-objective candidate and uses more Zone J supply.
- v2 is preferred after correcting interface normalization with fixed interface scales.
- The corrected fixed-scale objective avoids over-weighting missing P-32 limits with a 500 MW fallback.
- Current recommendation: use v2 as the fixed-scale S4 candidate; keep v1 as a Zone-J-heavy sensitivity.

Current Tie-Line / Interface Mismatch

Interface	Fixed scale	Mean abs. residual	Max abs. residual
Total East proxy	2905 MW	938 MW	1800 MW
Central East	1393 MW	651 MW	751 MW
Moses South	1125 MW	459 MW	605 MW
UPNY-ConEd	2999 MW	437 MW	1655 MW
Dunwoodie South	1936 MW	411 MW	1413 MW

Interpretation

The remaining problem is not raw ACOPF feasibility or branch overload. It is behavioral mismatch: the reduced tie-line and generation-equivalent model still does not reproduce all scaled P-32 interface patterns.

Why Mismatch Remains

- P-32 interface flows are aggregate system outcomes, not individual branch ratings.
- S4 still uses reduced proxy corridors and equivalent generators instead of detailed NYISO internal topology.
- Zone J vs G/K/F/E equivalent dispatch shifts the burden between local supply and upstream imports.
- Total East remains sensitive to A/B/C participation, F/G supply, and Gilboa–Leeds / downstate-spine reactance.
- UPNY–ConEd and Dunwoodie South remain sensitive to how much NYC load is served locally versus through lower-Hudson paths.

The mismatch is now a calibration problem, not an overvoltage rescue problem.

Next Work Items

❶ **Refine the fixed-scale S4 candidate.**

Sweep near v2: Zone J and S4 equivalent costs around 180–220, A/B/C cap factor around 0.75–0.85, and X at 1.0 / 0.75.

❷ **Report dispatch plausibility.**

Track Zone J, F, G, K, E, and A/B/C generation by scenario so equivalents do not become hidden slack resources.

❸ **Add voltage-deviation diagnostics.**

Voltage penalty reduced bound hits in v1 tests, but did not eliminate them.

❹ **Prototype exact AC interface objective later.**

Do not fake interface flow penalties as linear generalized costs; exact AC branch-flow penalties should use nonlinear-cost callbacks.

❺ **Keep holdout scenarios.**

Candidate selection should be tested on scenarios not used for tuning.

Takeaways

- The old relaxed-voltage solution identified a true modeling issue: public-load cases could escape to system-wide 2.0 pu voltages.
- S2 added missing downstate delivery structure but was not enough on its own.
- S3 fixed the main feasibility problem by adding bounded NYC / Zone J active-power controllability.
- S4 aligned generation capability and participation using Gold Book zonal patterns, giving a much stronger calibration candidate.
- The present limitation is tie-line and interface-flow mismatch, especially Total East, not ACOPF convergence.

Current status

Feasible and materially improved; still candidate-level interface calibration.

Local Artifacts

Artifact	Role
npcc_ny_lite_s4_cost_calibration_candidate_v2.m	Fixed-scale S4 candidate case.
s4_cost_penalty_fixed_scale_rescore_summary.csv	Corrected fixed-scale candidate ranking.
s4_candidate_validation_combined_20260703_212010.zip	Combined S4 candidate validation archive.
nyiso_interface_objective_scales.csv	Fixed interface normalization scales.
ny_only_case_d_vmax2_overvoltage_buses.csv	Relaxed-voltage overvoltage evidence.

All results shown here are from local MATPOWER/IPOPT runs and project CSV outputs.